

Limited conifer regeneration following wildfires in dry ponderosa pine forests of the Colorado Front Range

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Abstract. In recent years, increased wildfire activity and climate change have raised concern among scientists and land managers regarding current and future vegetation patterns in post-burn landscapes. We surveyed conifer regeneration 8–15 years after fire in six burn areas in the lower montane zone of the Colorado Front Range. We sampled across a broad range of elevations, aspects, and fire severities and found that densities of ponderosa pine (*Pinus ponderosa*) and Douglas-fir (*Pseudotsuga menziesii*) are generally low, although areas of abundant regeneration do occur. Conifer regeneration was most limited in xeric settings, including more southerly aspects and elevations closer to lower treeline. Additionally, fewer juvenile conifers occurred at greater distances from mature, live trees indicating that seed source as well as topoclimatic setting limits post-fire tree regeneration. Projecting the extent of future forest cover is uncertain due to the possibility of future pulses of tree establishment and unknown depletion rates of existing seedling populations. However, current patterns of post-fire seedling establishment suggest that vegetation composition and structure may differ notably from historic patterns and that lower density stands and even non-forested communities may persist in some areas of these burns long after fire, especially in xeric settings or where no nearby seed source remains.

Key words: climate change; Colorado Front Range; conifer regeneration; Douglas-fir; ecotone; fire severity; *Pinus ponderosa*; ponderosa pine; *Pseudotsuga menziesii*; resilience; tree establishment; wildfire.

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INTRODUCTION

An important concept in ecology is that of resilience, or the ability of an ecosystem to recover to a similar state following disturbance (Holling 1973, Gunderson and Holling 2002). In the western United States, concern has grown among researchers and land managers that recently burned landscapes may be exhibiting lower resiliency to fire. Grasslands or shrublands now dominate in some previously forested areas, at least in portions of burns where seed availability is low (Savage and Mast 2005, Keyser et al. 2008, Roccaforte et al. 2012, Dodson and Root

2013, Ouzts et al. 2015). More research is needed to determine whether limited post-fire conifer regeneration is a widespread phenomenon in the western United States, and whether climate change, increased fire severity, limited time since fire, or a combination of these and other factors explain observed patterns.

A recent synthesis of climate change in Colorado (Lukas et al. 2014) reported that statewide annual average temperatures have risen by 1.1°C over the last 30 years and by 1.4°C over the past 50 years. A similar study focused on the CFR also documented warming in recent decades, including in the lower montane zone where the

present study is situated (McGuire et al. 2012). Statewide, additional increases of 1.4–3.6°C are expected by 2050 (Lukas et al. 2014). In contrast, changes in precipitation exhibit no clear trend statewide and future precipitation patterns remain difficult to predict (Lukas et al. 2014). Given the strong association between climate variability and vegetation patterns (Prentice 1986, Briffa et al. 2004), altered temperature and/or precipitation regimes are expected to result in significant changes in forest composition, structure, and function. These changes are likely to occur most rapidly when facilitated by disturbances such as fire, lethal insect outbreak, or other events that result in widespread mortality of mature trees (Vose et al. 2016). In these situations, abundant conifer regeneration is needed to promote conditions similar to those prior to disturbance, yet successful regeneration may not occur if climate conditions are no longer suitable (Hogg and Schwarz 1997, Spittlehouse and Stewart 2003). Thus, fires may facilitate long-lasting changes in vegetation patterns under a persistent trend toward a warmer or drier climate. In the case of dry ponderosa pine (*Pinus ponderosa*) forests of the western United States, processes such as conifer seed production, germination, and subsequent establishment and survival are associated with specific climate requirements (Schubert 1974, White 1985, Savage et al. 1996, 2013, League and Veblen 2006, Feddema et al. 2013, Rother 2015, Rother et al. 2015). Warmer temperatures and associated drought are expected to inhibit conifer regeneration, at least where trees are growing near their distributional limit such as near lower treeline.

In addition to climate change, increased fire severity has also been identified as a possible driver of limited conifer regeneration after fire. It has been suggested that fire suppression has caused coniferous forests of the western United States to be susceptible to uncharacteristically severe fire (Covington 2000, Williams 2013), but these types of generalizations should be examined for particular landscapes and along elevation and moisture gradients within specific landscapes. In ponderosa pine forests of the Colorado Front Range (CFR), the historic fire regime was mixed severity, meaning that fire effects were varied both within stands and across the landscape and included

low-, moderate- and high-severity fire (Kaufmann et al. 2006, Sherriff and Veblen 2006, 2007). While historic fire regimes of exclusively low-severity fire were most common at the lowest elevations near the ecotone with the Plains grasslands (Sherriff et al. 2014), reconstructed fire frequencies, severities, and post-fire recovery indicate that higher severity fires (e.g., killing at least 70% of canopy trees) also occurred even at low elevations (Veblen and Lorenz 1986, Kaufmann et al. 2006, Sherriff and Veblen 2006). Although high-severity fire per se does not preclude successful ponderosa pine regeneration, large patches of high-severity fire are known to limit regeneration by leaving behind fewer seed trees (Bonnet et al. 2005, Haire and McGarigal 2010). Additionally, edaphic changes as well as microclimatic effects of blackened soil and absence of vegetation may result in altered microclimate conditions such as higher daily temperature ranges and reduced soil moisture (Ulery and Graham 1993, Wondafrash et al. 2005, Montes-Helu et al. 2009).

In the present study, we examined patterns of post-fire conifer regeneration in low-elevation, ponderosa pine forests of the CFR. Our primary research objectives were to: (1) quantify post-fire conifer establishment and survival and (2) examine the variability of juvenile conifer regeneration in relation to site factors such as fire severity, competition with herbaceous and woody species, distance to seed source, and topographic variables including elevation and aspect. Given the sensitivity of the regeneration success of ponderosa pine and Douglas-fir (*Pseudotsuga menziesii*) to climate variability (Schubert 1974, White 1985, Savage et al. 1996, 2013, League and Veblen 2006, Feddema et al. 2013, Rother 2015, Rother et al. 2015) and that air temperature has increased in recent years (McGuire et al. 2012, Lukas et al. 2014), we hypothesized that conifer regeneration may be limited across the study area, especially in hotter, drier settings such as at lower elevations and on south-facing aspects. We also expected that high-severity burning would inhibit post-fire tree establishment by reducing seed availability, given that seeds of both ponderosa pine and Douglas-fir are wind-dispersed over relatively short distances (Bonnet et al. 2005, Shatford et al. 2007, Haire and McGarigal 2010).

METHODS

Study area

The study area is along the eastern slope of the CFR, in the lower montane zone (Kaufmann et al. 2006), where dry ponderosa pine forests occur (Fig. 1). Mean maximum January temperature is approximately 4.1°C, mean maximum July temperature is approximately 26.4°C, and annual mean precipitation is approximately 42.2 cm (Bailey COOP Station, 2360 m, period of record: 1901–2013). Climate stations along an elevation gradient centrally located in our study area record warming trends over the periods 1953–2008 and 1989–2008, largely through increases in maximum temperatures (McGuire et al. 2012). Recent decades show an increase in maximum temperatures at the elevations of our sample sites and increased warming during July (McGuire et al. 2012).

Forest vegetation patterns in the CFR are strongly influenced by moisture variability related

to both elevation and aspect (Peet 1981). At the lower elevational range of the lower montane zone, ponderosa pine is dominant and forms relatively open stands. With increasing elevation and moisture availability, stand density increases and Douglas-fir is often present or co-dominant (Kaufmann et al. 2006). The elevational range of the lower montane zone varies with latitude and microsite conditions, ranging from approximately 1675 to 2285 m (5500–7500 ft) in the northern CFR and from approximately 1980 to 2590 m (6500–8500 ft) in the southern CFR (Kaufmann et al. 2006). These forests are characterized by frequent disturbances by fire and insect attack and are typified by varying stand ages and species compositions (Peet 1981).

Site selection and field sampling methods

Potential burn areas to study were identified using GIS data layers of recent fires from the Monitoring Trends in Burn Severity Program (MTBS). MTBS includes fire perimeter and severity data for all U.S. wildfires that have occurred since 1984, except for small fires (<200–400 ha, depending on the region of the country). MTBS produces fire-severity data using the differenced Normalized Burn Ratio (dNBR), calculated from satellite imagery from LANDSAT. For this study, we generated a list of all wildfires of over 400 hectares that occurred mostly or entirely within the lower montane zone of the CFR between 1984 and 2003. Of the nine fires in this subset, we selected six fires to serve as study sites (Figs. 1 and 2; Appendix S1): the Buffalo Creek fire of 1996 (BC), the Bobcat Gulch fire of 2000 (BG), the Hayman fire of 2002 (HY), the High Meadows fire of 2000 (HI), the Overland fire of 2003 (OL), and the Walker Ranch fire of 2000 (WR). The other three fires on the initial list were excluded due to limited access related to isolation and/or land ownership issues. We only included MTBS fires that occurred prior to 2004 because we wanted time since fire to be sufficient enough that post-fire conifer establishment should be well underway. Time since fire ranged from 8 to 15 years at time of survey, which is longer than average intervals of no or low seed production for ponderosa pine (Shepperd et al. 2006, Mooney et al. 2011). Fires ranged in size from approximately 400–52,000 ha, and all fires included a mixture of low- to high-severity

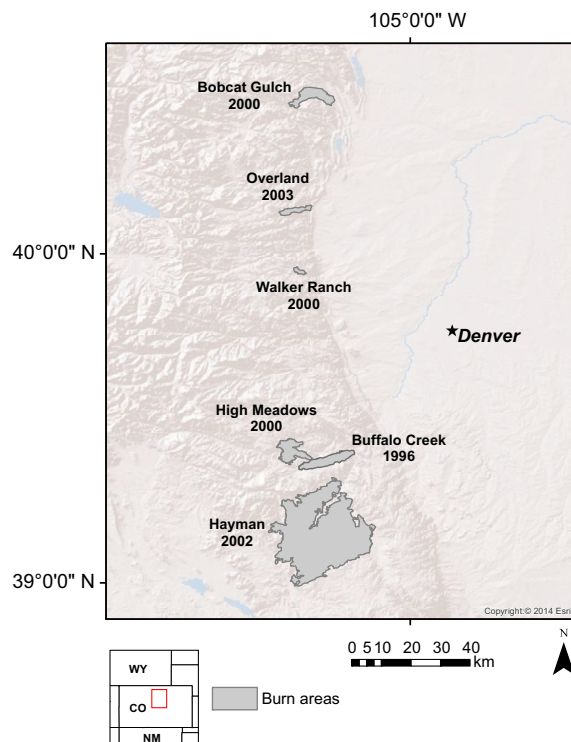
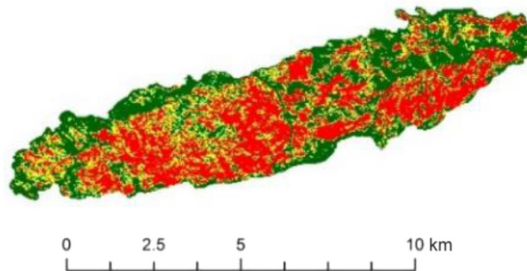


Fig. 1. Study area map including the name and year of the six burn areas.

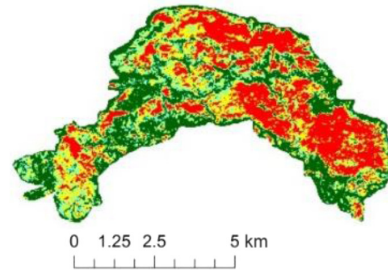
(a) Buffalo Creek, May 1996: 3962 ha

High = 38%
Moderate = 21%
Low/Unburned = 41%



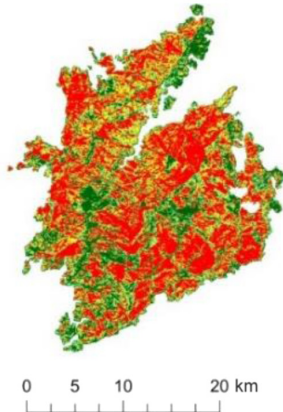
(b) Bobcat Gulch, June 2000: 3694 ha

High = 30%
Moderate = 23%
Low/Unburned = 47%



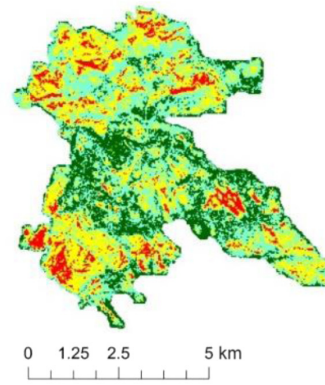
(c) Hayman, June 2002: 52,368 ha

High = 43%
Moderate = 22%
Low/Unburned = 34%



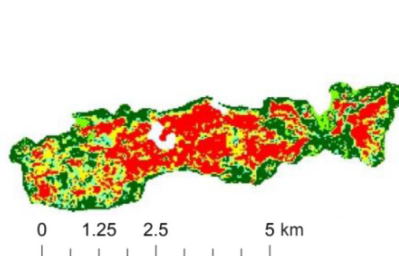
(d) High Meadows, June 2000: 3889 ha

High = 8%
Moderate = 32%
Low/Unburned = 59%



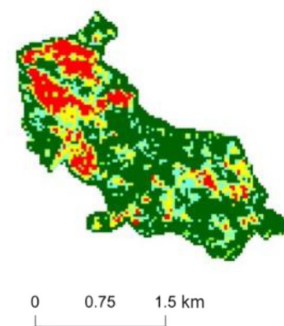
(e) Overland, October 2003: 1308 ha

High = 35%
Moderate = 21%
Low/Unburned = 39%



(f) Walker Ranch, September 2000: 404 ha

High = 13%
Moderate = 16%
Low/Unburned = 70%



patches within their perimeters (Fig. 2). The variability of vegetation cover with aspect and along elevation gradients across the study area is well documented (Peet 1981, Kaufmann et al. 2006,

Keith et al. 2010) and was used to guide the design of vegetation sampling (Appendix S1). The diversity of site characteristics within and between fires is advantageous because it allows

Fig. 2. Fire-severity data for the six mixed-severity burn areas included in the study. The header for each fire includes the fire name, ignition date, and total size. Data are remotely sensed and are from the Monitoring Trends in Burn Severity Program (MTBS). Red areas indicate high severity, yellow areas indicate moderate severity, dark green areas indicate unburned to low severity, light green areas indicate low severity, bright green areas indicate increased greenness, and white areas indicate non-processing area mask. For the provided percentages of each fire-severity class above, low severity (light green) is combined with the unburned to low severity category (dark green).

for comparison of post-fire vegetation patterns across varying settings and enables inferences across relatively broad spatial and temporal scales.

To survey current vegetation patterns in each of the six burn areas, we used belt transects stratified by aspect and fire severity. Each belt transect (hereafter, "plot") measured 2×50 m and was divided into 10 quadrats (of 2×5 m each) to facilitate more accurate assessment of vegetation attributes (e.g., cover estimates). Sampling was stratified into a total of two aspect settings (north and south) and three fire-severity settings (low, moderate, and high). North-facing aspects were defined as ranging from NW (315°) to NE (45°), while south-facing aspects ranged from SW (225°) to SE (135°). We determined fire-severity classes by estimating percentage canopy tree mortality for the stand in which the plot was situated, using three classes: low (0–20%), moderate (21–80%), and high (81–100%). Thus, our survey work concentrated on six general fire-severity/aspect settings: high-severity/north-facing (HN), high-severity/south-facing (HS), moderate-severity/north-facing (MN), moderate-severity/south-facing (MS), low-severity/north-facing (LN), and low-severity/south-facing (LS). In each burn, we collected 5–13 replicates per setting, for a total of 41–60 plots per burn. This resulted in 302 plots for the entire study area. Although the boundaries between aspect and fire-severity classes were subjectively set and are necessarily broad, we also recorded the precise aspect (degrees) and canopy mortality (%) for each plot, from the plot center. Suitable areas for sampling within each burn area were first identified by viewing KMZ files of MTBS data in Google Earth. Specifically, we identified locations that were relatively uniform in terms of fire severity and aspect. Then, in the field, we randomly situated midpoints for

plots and then extended the plots parallel to the slope contour.

At the center of each plot, data collected included elevation (m), aspect ($^\circ$), canopy mortality (%), slope gradient ($^\circ$), and GPS location. For comparison between sites, elevation was later adjusted to account for differences in latitude. The northernmost site (BG) was adjusted by adding 500 m to the elevation, while the southernmost sites (HI, BC, HA) were adjusted by subtracting 500 m from the elevation. These adjustments were based on previous work defining the elevational ranges of the lower montane zone for the southern, central, and northern Front Range (Kaufmann et al. 2006). In each quadrat, we collected data on the number, height, and species of all post-fire juvenile conifers, the distance to nearest seed source (m), canopy cover (%), and various substrate and vegetation cover data (Table 1). The substrate and vegetation cover data were collected using a modified Braun-Blanquet cover type approach. Cover classes were defined as: 1 = $<1\%$; 2 = 1–4.99%, 3 = 5–24.99%, 4 = 25–49.99%, 5 = 50–74.99%, 6 = 75–100. Post-fire juvenile conifers were defined as trees of height <1.4 m. Because it is not possible to be certain that an individual tree established pre- or post-fire, this may have led to slightly higher post-fire juvenile conifer counts, especially in low-severity settings where small trees could have survived the fire. However, the majority of juvenile conifer trees we encountered were less than 0.3 m in height, which by size allows us to estimate their ages as within the post-fire time period (Veblen and Lorenz 1986, Kaufmann et al. 2000, Sherrieff and Veblen 2006). Lastly, to allow for comparisons of pre- vs. post-fire conifer densities, we counted trees with diameter at breast height (dbh) > 15 cm including both: (1) live mature trees and snags with at least 50% of their

Table 1. Summary of predictor variables of conifer regeneration used in the Random Forest (RF) analyses.

Variable name and expected relationship (+ or –)	Variable definition	Significance for conifer regeneration
Dist. to Seed Source (–)	Distance (m) from nearest mature, live ponderosa pine or Douglas-fir	Both ponderosa pine and Douglas-fir are wind-dispersed over short distances.
Adjusted Elevation (+)	Elevation (m) adjusted based on latitude (see text for details)	Moisture availability increases and temperature decreases with increasing elevation in the study area.
Aspect Variable (–)	Absolute number of degrees away from north (0°)	North-facing slopes are cooler and more mesic than south-facing slopes.
Canopy Mortality (+ or –)	Canopy mortality (%), estimated based on the area immediately surrounding the plot (~25 m radius of plot center)	Canopy mortality is a measure of fire severity, with higher canopy mortality corresponding to higher fire severity. Low canopy mortality may create conditions that are too shady for conifer regeneration. High canopy mortality may create conditions that are too hot and dry.
Canopy Cover (+ or –)	Canopy cover (%)	Canopy cover corresponds to the amount of sunlight that reaches the forest floor. Low canopy cover may create conditions that are too hot and dry for conifer regeneration. High canopy cover may create conditions that are too shady.
Slope Gradient (–)	Slope gradient (°)	Steeper slopes retain less moisture and may be less suitable for conifer regeneration.
Bare Soil (+ or –)	Cover class of bare soil	Bare soil provides a competition-free space for regeneration, but can also indicate hotter, drier microclimate conditions.
Litter (+ or –)	Cover class of litter	Cover by litter is an indicator of crown cover.
C.W.D. (+)	Cover class of coarse woody debris	Coarse woody debris can provide a “safe site” for germination by increasing soil moisture and decreasing air/soil temperature.
Graminoids (–)	Cover class of graminoids	High graminoid cover can create competition for conifer regeneration.
Forbs (–)	Cover class of forbs	High forb cover can create competition for conifer regeneration.
Shrubs (–)	Cover class of woody understory vegetation	High shrub cover can create competition for conifer regeneration.
Rocky (–)	Cover class of exposed rock	Solid rock is unsuitable for conifer regeneration.

Note: Cover classes were defined as: 1: <1%; 2: 1–4.99%; 3: 5–24.99%; 4: 25–49.99%; 5: 50–74.99%; 6: 75–100%.

tree bole located in the plot and (2) fallen trees >15 cm dbh that had been rooted in the plot and appeared to have been killed by the fire.

Data analyses

We used the number of juvenile conifers in each plot to extrapolate the density (stems/ha) of juvenile conifers in each burn study area. We also calculated a pre-fire and post-fire total tree density for each plot and compared the frequency distribution of pre- vs. post-fire data using Mann–Whitney *U* tests. Pre-fire densities were based on the number of live mature trees, snags, and fallen trees, while post-fire densities were based on live mature and live juvenile trees. To examine the variability of juvenile conifer regeneration in relation to site factors (Table 1), we used Random Forests (RF) (Breiman 2001) to develop classification models, using the package randomForest (Liaw and Wiener 2002) in R

(R Development Core Team). We developed classification models of the (1) probability of juvenile conifer presence or absence and (2) the probability that plots were stocked or unstocked with conifer seedlings. For the stocked/unstocked model, we used standards defined by the National Forest Management Act. We specifically relied on desired stocking levels of seedlings in ponderosa pine forests described in the Forest Plan for the Arapaho National Forest (USFS 2012) to characterize our plots (i.e., each 2 × 50 m plot) as either stocked or unstocked. Stocked plots met a threshold of at least 370 trees/ha, whereas unstocked plots had fewer than 370 trees/ha. These stocking levels are generally intended for use in a silvicultural context, but are relevant for our study in that they provide a conservative threshold above which land managers should feel confident that seedling densities are sufficiently high for forest recovery.

We chose RF as a means of determining important predictors of post-fire conifer regeneration because it is known to work well with ecological data that are complex and nonparametric (Cutler et al. 2007). RF is an extension of classification and regression tree (CART) analysis, whereby trees are constructed by repeatedly dividing the data into two mutually exclusive groups (Breiman 2001). RF has been recognized as effectively identifying important ecological relationships. Through RF, many trees are fit to the data and then later combined. The output allows the user to identify the variables that are most important for prediction, and has frequently been used to select a subset of variables for input into subsequent analyses (e.g., Thompson and Spies 2009, Hart et al. 2014). We specifically used the mean decrease in accuracy statistic to select the top three most important variables. This statistic is a measure of how much the inclusion of the variable reduces classification error. The RF analysis includes the out-of-bag (OOB) error estimate. A lower OOB error estimate indicates a higher accuracy of classification (e.g., if the OOB error estimate = 30%, accurate classification occurred 70% of the time). Our analysis included data from all six of the study areas (all 302 plots). We used the default setting for the number of variables randomly sampled as candidates at each split, which in our case was 3. The number of trees grown in each iteration was set to 500. Following our RF analyses, the top three predictor variables were then used to construct classification trees in the rpart package (Therneau et al. 2015) in R. To avoid overfitting the data, the tree was pruned using a complexity parameter that minimized the cross-validated error. Classification trees are useful to identifying thresholds that are important in separating data and can reveal complex, nonlinear relationships.

RESULTS

We collected data in 41–60 plots per burn, for a total of 302 plots. A relatively even number of plots were collected in each of the six settings of aspect and burn severity. Most plots sampled contained no conifer seedlings ($n = 178$, 59%, Fig. 3), and an even larger percentage of plots ($n = 250$, 83%, Fig. 4) were considered unstocked

(i.e., the density of seedlings was below 370 stems/ha). Depending on the fire, only 2–38% of plots were stocked. The density data are not normally distributed, but mean density of juvenile conifers for ponderosa pine and Douglas-fir combined ranged from approximately 40 to 1420 trees/ha; all sites besides HY had mean densities of only approximately 40–240 trees/ha (Fig. 3). At the HY burn, many of the post-juvenile conifers were small (<10 cm height), and mean densities were notably lower (mean = 619 trees/ha) when the smallest height class (i.e., 0–10 cm) was excluded (Appendix S2). Average post-fire tree density based on mature and juvenile trees combined is significantly lower ($P < 0.001$) than pre-fire density at all sites except for HY (Fig. 5). In terms of tree species regenerating, we observed almost only ponderosa pine and Douglas-fir (more ponderosa pine than Douglas-fir, Appendix S2). A small number of aspen (*Populus tremuloides*), lodgepole pine (*Pinus contorta*), and Rocky Mountain Juniper (*Juniperus scopulorum*) were also observed, but due to low counts across the study area, these species were not included in the analyses. Substrate and vegetation cover were highly variable, but in most plots ($n = 231$, 76.4%), cover by graminoids was equal or higher than that by shrubs or forbs. Only a small portion of plots ($n = 47$, 15.6%) contained a notable amount of bare soil (at least 5% cover).

Conifer regeneration was generally low across all fire-severity settings. Although in some burn areas, conifer densities were somewhat higher in one or two of the fire-severity settings, no clear patterns were observed across the six burn areas; low-severity fire did not consistently correspond with higher seedling densities, and high-severity fire did not consistently correspond to lower seedling densities (Fig. 6). However, in the field, we observed that exceptionally large patches (e.g., diameter > 250 m) of very high-severity fire (canopy mortality > 95%) were typically devoid of juvenile conifers.

Exploratory analysis of the data (i.e., graphically displaying the data) suggested that distance to seed source, elevation, and aspect might be important for understanding patterns of conifer regeneration. Regeneration was most abundant within close proximity (<50 m) of one or more remnant trees (i.e., the seed source), at higher

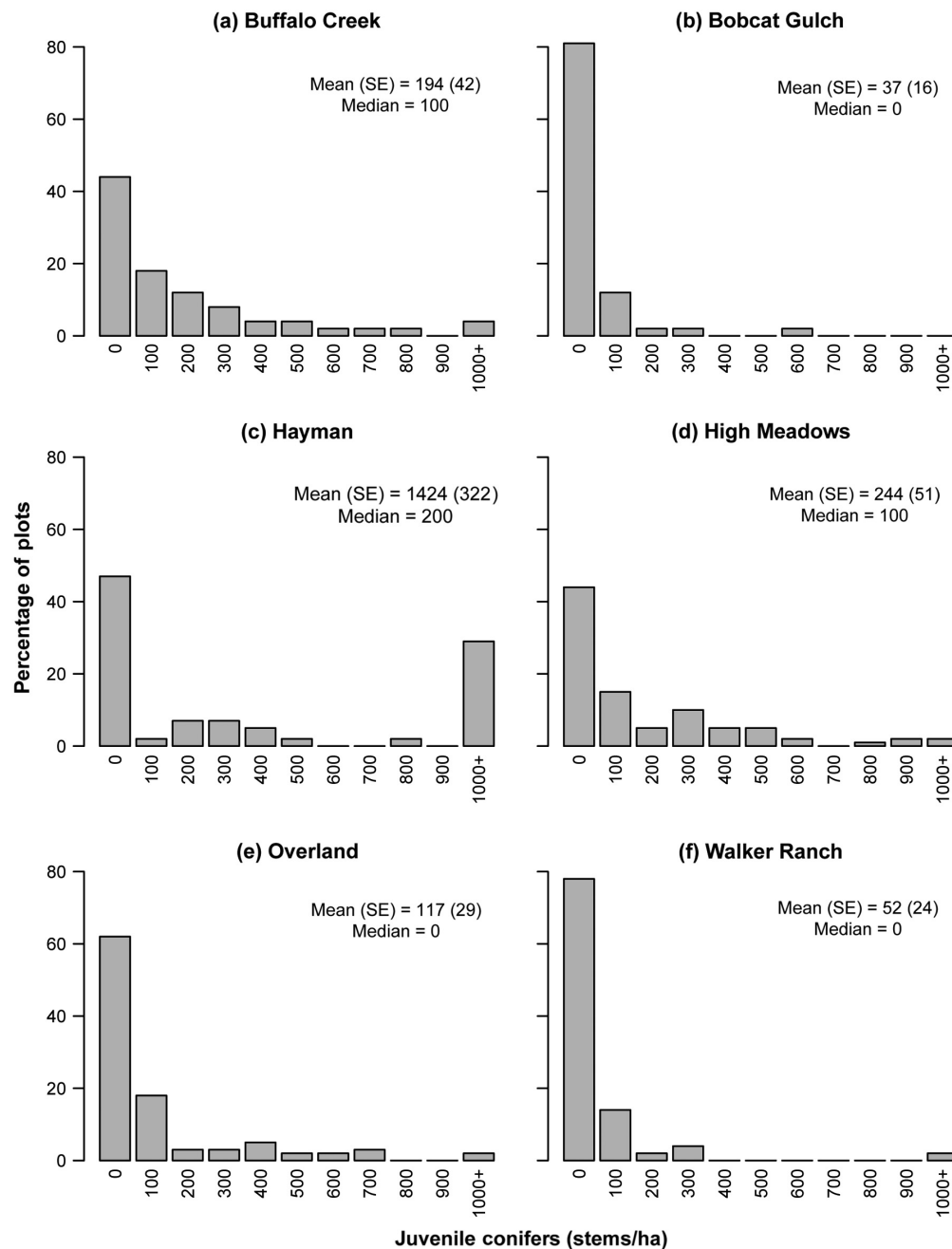


Fig. 3. Percentage of plots (each 2×50 m) in varying classes of juvenile post-fire conifer densities (stems/ha). At all sites, more plots were in the 0 stems/ha class than any other class.

elevations, and on north-facing slopes. RF analysis supported the exploratory analysis. For our presence/absence model, the top three predictor variables in order of descending importance were as follows: adjusted elevation, the aspect

variable, and distance to seed source (Fig. 7a). For our stocked/unstocked model, the top three predictor variables in order of descending importance were as follows: adjusted elevation, distance to seed source, and the aspect variable

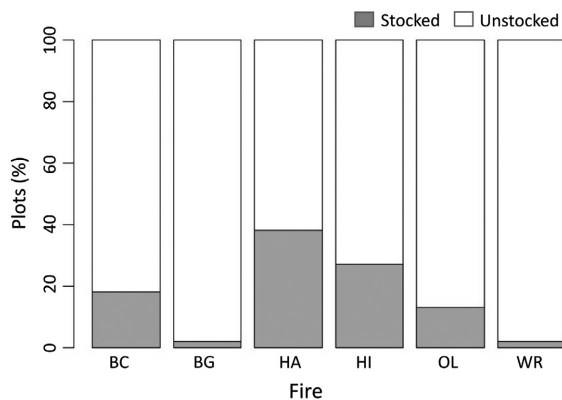


Fig. 4. Percentage of plots that were stocked (tree density ≥ 370 stems/ha) vs. unstocked (tree density < 370 stems/ha) at time of survey in each of the six burn areas. BC = Buffalo Creek, BG = Bobcat Gulch, HA = Hayman, HI = High Meadows, OL = Overland, WR = Walker Ranch.

(Fig. 7b). Canopy mortality (an indicator of fire severity) and crown cover were lower in importance, as were the substrate and vegetation cover variables. The general association between adjusted elevation and probability of conifer presence (Fig. 7a) or stocking (Fig. 7b) was positive (increased elevation = higher probability of conifer seedling presence and stocking), while relationships with distance to seed source and the aspect variable were negative (greater distance from seed source and more southerly aspect = lower probability of seedling presence and stocking), although relationships were not linear. For the presence/absence model, the OOB error estimate was 28.5%, indicating that accurate classification occurred 71.5% of the time. For the stocked/unstocked model, the OOB error estimate was 15.6%, indicating that accurate classification occurred 84.4% of the time. Our RF analysis allowed for the construction of classification trees that included the predicted condition (juvenile conifer presence or absence; stocked or unstocked) and the probability that the predicted response will occur given the path leading to the node (Fig. 8). We found that below a threshold of 2368 m for adjusted elevation, plots were classified as absent (i.e., containing zero seedlings) and unstocked (Fig. 8a, b). Above 2368 m, plots were predicted to be unstocked if distance to seed source was greater than or equal to 10 m (Fig. 8b).

DISCUSSION

Our findings indicate that in the burn areas we sampled in the lower montane zone of the CFR 8–15 years after fire, ponderosa pine and Douglas-fir post-fire regeneration is constrained by elevation, aspect, and proximity to seed sources. The majority of plots contained zero post-fire juvenile seedlings, and most areas do not meet the Arapaho National Forest stocking threshold of 370 juvenile trees/ha (USFS 2012). A small proportion of plots were characterized by high densities of seedlings (sometimes exceeding 1000 seedlings/ha), especially in the HY burn. We also found that average post-fire tree density based on the number of mature and juvenile trees combined is significantly lower than pre-fire density at all sites except the HY burn.

With regard to where post-fire juvenile conifers are most likely to occur, topoclimate (i.e., local climate conditions driven by characteristics of the terrain including aspect and elevation) and distance to seed source were found to be the most important predictors of conifer regeneration, using both presence/absence and adequate stocking as dependent variables in the RF analyses. Low numbers of seedlings in hotter and drier settings including elevations closer to lower treeline and south-facing aspects are consistent with previous research showing that the establishment and subsequent survival of both ponderosa pine and Douglas-fir depends on adequate moisture levels (Schubert 1974, White 1985, Savage et al. 1996, 2013, League and Veblen 2006, Feddema et al. 2013, Rother 2015, Rother et al. 2015). Similarly, an expectation of reduced regeneration success following recent fires is congruent with the regional trend in the CFR toward hotter and drier climatic conditions over the past several decades (McGuire et al. 2012, Lukas et al. 2014) and with climate niche models predicting an approximately 50% decline in the climate space of ponderosa pine by 2060 (Rehfeldt et al. 2014). Our findings are consistent with others that demonstrate the importance of topoclimate and/or post-fire climate conditions (e.g., drought after fire) in driving tree regeneration patterns, including other studies in the Southwest (Feddema et al. 2013, Savage et al. 2013), northern Rockies (Kemp et al. 2015, Harvey et al. 2016), CFR (Chambers et al. 2016), and Pacific Northwest (Dodson and Root 2013). In our

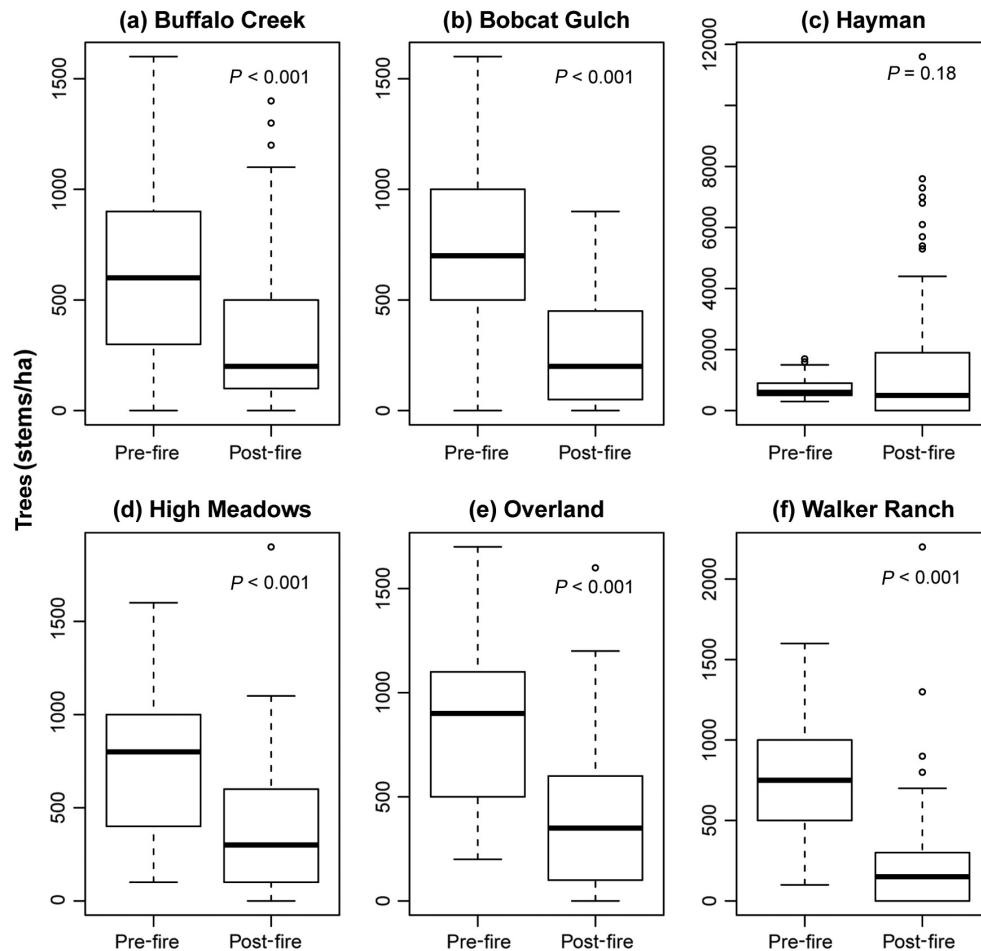


Fig. 5. Pre-fire vs. post-fire tree densities in each of the burn areas. Pre-fire densities were based on the number of live mature trees, snags, and fallen trees in each plot, while post-fire densities were based on live mature and live juvenile trees. Mann–Whitney U tests were used to determine whether the frequency distribution of pre-fire vs. post-fire density data was significantly different, and P -values are displayed within each chart. The thick black line inside the box indicates the median, the lines at the outer edges of the box indicate the upper and lower quartiles, and the lines at the end of vertical dashed lines indicate the maximum and minimum values. The dots indicate any outliers.

study and most of the cited studies, distance to seed source was also found to be an important predictor of conifer regeneration. Although this variable is related to fire severity, it is important to note that short distances to seed source often occur in high-severity areas when mortality is lower than 100% and that distance to seed source in lower-severity areas can also be quite variable. Many studies in other western forests have similarly identified distance to seed source to be important in explaining patterns of tree regeneration (e.g., Bonnet et al. 2005, Savage and Mast

2005, Keyser et al. 2008, Haire and McGarigal 2010, Roccaforte et al. 2012, Dodson and Root 2013, Ouzts et al. 2015, Chambers et al. 2016). Although our RF analysis identified elevation, aspect, and distance to seed source to be important predictors of conifer regeneration patterns, not all suitable areas for regeneration (i.e., high-elevation, north-facing slopes close to seed source) are characterized by adequate or abundant seedling densities, and conversely, seedlings are also not completely absent in all unsuitable areas (i.e., low-elevation, south-facing slopes that are far

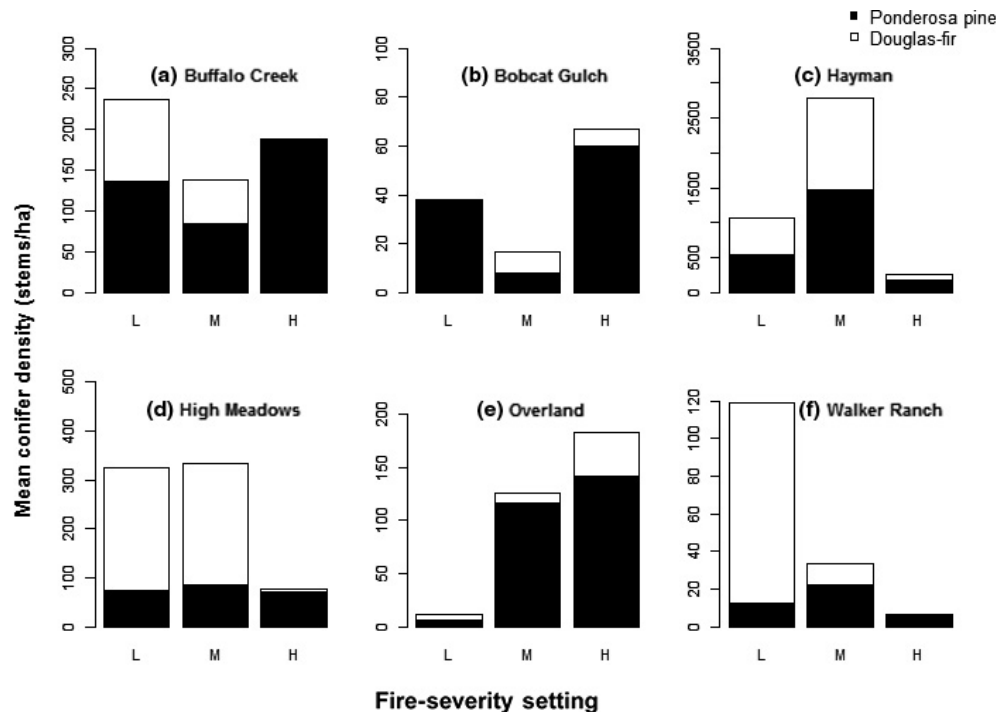


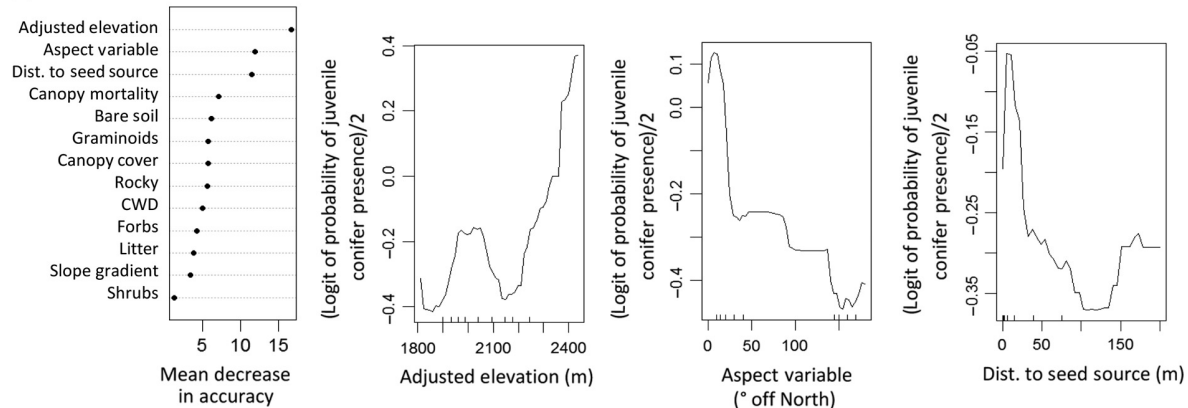
Fig. 6. Mean conifer density (stems/ha) occurring in low-, moderate-, and high-severity settings. L = low severity, M = moderate severity, and H = high severity. Fire-severity classes were determined by estimating percentage canopy tree mortality for the stand in which the plot was situated, using three classes: low (0–20%), moderate (21–80%), and high (81–100%).

from a seed source). There is therefore variability in seedling presence not fully accounted for by the predictor variables employed in this study.

It is difficult to make predictions about future forest conditions from our data given uncertainties about future seedling establishment and seedling population depletion rates. Thus, we stress that our findings are based on documenting a lack or low density of seedling establishment over a period of only 8–15 years following fire and can only be used cautiously to project future conditions. Previous research indicates that where dense stands of ponderosa pine and Douglas-fir burned in the 19th century, post-fire regeneration generally initiated quickly in the lower montane zone of the CFR (Veblen and Lorenz 1986, Kaufmann et al. 2000, Ehle and Baker 2003, Sherriff and Veblen 2006). However, the duration of post-fire regeneration periods undoubtedly varies with seed availability and site conditions affecting rates of initial seedling establishment and time required for attainment of relatively shaded understories

unsuitable for additional establishment of shade-intolerant trees. Several tree-ring-based studies have documented that post-fire establishment in ponderosa pine stands comparable to the forests burned in our study areas was historically concentrated soon after fire within a period of approximately 0–30 years (Veblen and Lorenz 1986, Kaufmann et al. 2000, Ehle and Baker 2003, Sherriff and Veblen 2006). In most cases, the post-fire pulse of regeneration initiated within the first decade after fire. Collectively, these previous studies suggest that the typical time window for post-fire tree establishment has not yet closed at our sites, but that 8–15 years after fire we should be observing at a minimum a pulse of initial seedling establishment. Although one study documented patches with no regeneration up to 150 years after fire (Kaufmann et al. 2000), exceptionally long windows of protracted post-fire seedling establishment are not the historical norm following the burning of dense ponderosa pine and Douglas-fir forests in the CFR.

(a) Presence/Absence



(b) Stocked/Unstocked

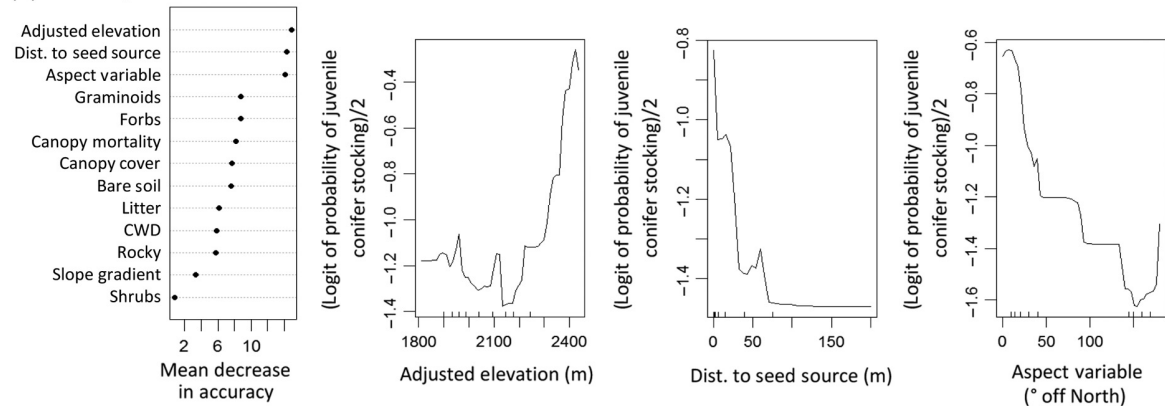


Fig. 7. Results from Random Forest (RF) analysis of (a) juvenile conifer presence/absence, and (b) juvenile conifer stocking, both for the combined dataset of all fires. Variable importance plots are on the far left of each figure and rank variables by mean decrease in accuracy. Mean decrease in accuracy is the normalized difference of the accuracy of the classification when the data for that variable are included vs. when they have been randomly permuted. Higher values indicate variables that were more important to the RF analysis. Partial dependence plots are to the right of the variable importance plots and show the dependence of the probability of juvenile conifer presence or adequate juvenile conifer stocking on one predictor after the effects of the other predictor variables are averaged out.

A limitation of comparing our findings to tree-ring-based studies is that total ages of large trees cannot be precisely determined with increment core data and thus are typically binned in 10–20 year bins. Additionally, because only origins of live trees at time of sampling can be determined, these tree-ring studies are unable to capture rates of seedling depletion.

A pattern that stands out strongly in our study is that the Hayman fire is a prominent exception to the pattern of scarce conifer regeneration evident at the five other burns. The Hayman fire

remains the largest fire in Colorado history in terms of area burned and is of special interest to land managers, researchers, and the public. We sampled only in lower montane portions of this burn, and thus, our findings do not apply to areas with elevations above approximately 2590 m. As the southernmost burn site, the area where the Hayman is situated experiences more summer monsoonal rains (i.e., higher precipitation amounts in July and August, Cheesman COOP Station). Given the importance of growing season moisture amounts for promoting seedling

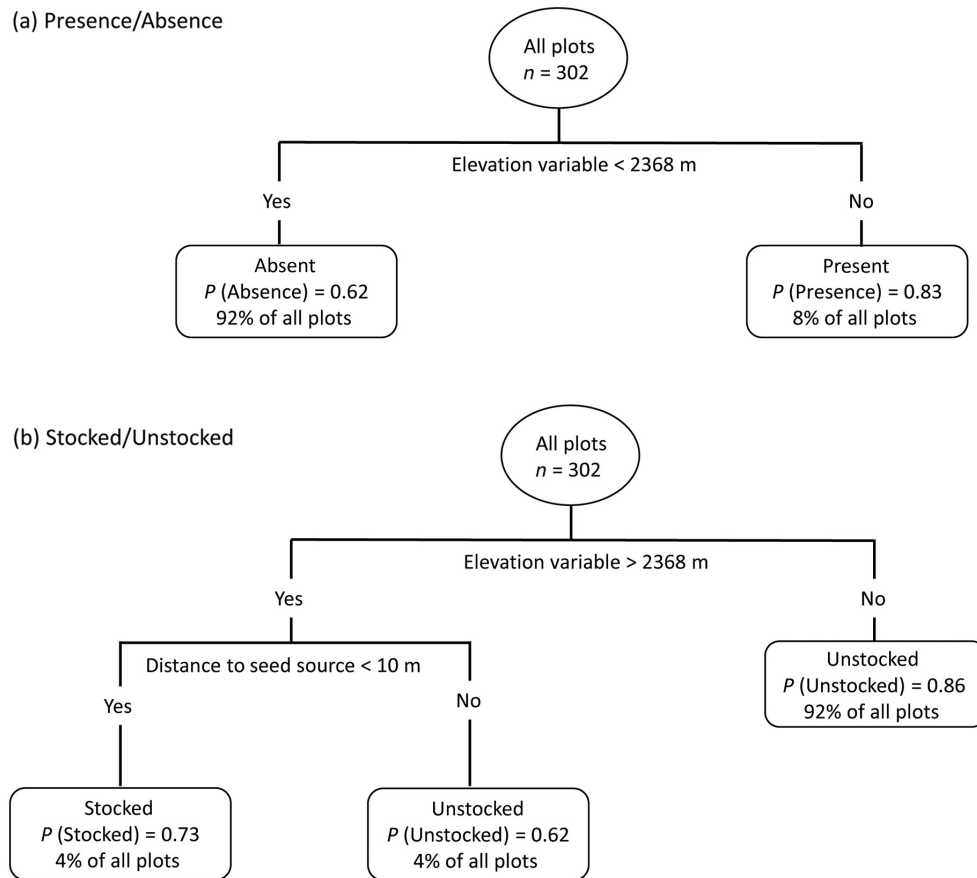


Fig. 8. Classification trees for determining (a) whether plots will have seedlings present, and (b) whether plots will be stocked. The trees were produced with the top three predictor variables selected using the mean decrease in accuracy statistic in the corresponding Random Forest (RF) analysis. At each terminal node, information presented includes (1) the predicted condition (absent/present or stocked/unstocked), (2) the probability (P) that the predicted response will occur given the path leading to the node, and (3) the percentage of observations.

establishment (Schubert 1974, White 1985, Savage et al. 1996, 2013, League and Veblen 2006, Feddema et al. 2013, Rother 2015, Rother et al. 2015), this difference in the timing and amount of precipitation may partially explain our finding. Although the Hayman is the only burn area with abundant regeneration, the nearby Buffalo Creek and High Meadows burn areas have higher average seedling densities than the more northerly burns. Additionally, the adjusted elevation of plots in the Hayman was the second highest among all fires (only Walker Ranch was higher), and this may also be relevant in understanding differences among sites.

Looking forward, episodes of abundant seedling establishment could result from one to

several years of above average moisture availability, which promotes establishment in open areas of ponderosa pine forests (i.e., not beneath forest canopies; League and Veblen 2006, Rother 2015). However, trends of increased temperature in Colorado (Lukas et al. 2014), including the CFR (McGuire et al. 2012), suggest that periods of time that are suitable for seedling establishment and subsequent survival are likely to occur less frequently in upcoming years, and thus, historic patterns may not be fully relevant in the context of changing climate. In the absence of abundant conifer regeneration at burned sites in the lower montane zone in future years, we expect that many areas that burned at high severity will persist as grasslands, with only sparse

remnant trees, particularly at lower elevations, more southerly aspects, and where seed availability is lowest. In contrast, in lower montane areas where canopy tree mortality was relatively low (i.e., in low-severity and moderate-severity burn areas), the forested vegetation type will likely persist into the future even without significant amounts of conifer regeneration. However, even these areas could face eventual transitions in vegetation type as mature trees senesce, or as additional disturbances (e.g., another fire, beetle outbreak) occur, unless regeneration patterns change.

Our data demonstrate that tree regeneration is limited in the burn areas we sampled, but this is not necessarily undesirable from a land management perspective. At the lowest elevations of where ponderosa pine forests occur in the CFR (and not at higher elevations), fire exclusion has resulted in higher tree densities than what is expected based on historic patterns (Kaufmann et al. 2006, Sherriff et al. 2014). Lower seedling densities may thus be interpreted as promoting stand conditions that are more consistent with restoration goals and may also have the added benefit of reducing the hazard of high-severity fire. Particularly in the wildland–urban interface, there is a convergence of ecological restoration and fire mitigation goals at the lowest elevations of the ponderosa pine zone in the CFR. Even if the average seedling densities we report seem adequate to some land managers, it is important to keep in mind that our density data were not normally distributed and that there are many areas that are completely devoid of seedlings. Large high-severity patches with virtually no surviving seed trees may be of particular concern. If land managers decide that current post-fire seedling densities are lower than desired, one option is to plant tree seedlings. However, the survival of planted seedlings is likely to be highly variable depending on the topoclimatic conditions identified in our study and in relation to subsequent regional climatic conditions. In a recent study of high-severity burn areas of Arizona and New Mexico, seedling survival was only 0–12% in three of the eight burns (Ouzts et al. 2015). This suggests that, consistent with our findings, seed availability is not the sole cause of limited tree regeneration.

Given that planting seedlings is costly and often unsuccessful and that a return to pre-fire conditions may not always be desired, accepting a shift from forest to non-forest vegetation or to lower density stands in some areas may be appropriate. If plantings are to occur, our findings suggest that land managers should target settings where topoclimate is less likely to inhibit subsequent survival such as on north-facing slopes and higher elevations.

CONCLUSIONS

We have documented low seedling densities 8–15 years post-fire in lower montane forests of the CFR. More time is needed to make confident predictions about future vegetation patterns, but our findings suggest that lower density stands and some areas of non-forested vegetation are likely to persist long into the future. Many studies in the Southwest have stressed the importance of seed limitations due to high fire severity in explaining lack of tree regeneration (Savage and Mast 2005, Haire and McGarigal 2010, Roccaforte et al. 2012). Although we also found distance to seed source to be important, we conclude that elevation and slope aspect (which strongly affect topoclimate) are important drivers of current regeneration patterns, especially in the context of climate change. As climate continues to warm, we anticipate increasingly limited post-fire regeneration in hotter/drier settings. Our study was focused on lower montane forests of the CFR, but we expect that similar changes are underway or imminent in other low-elevation forests where warmer climates may inhibit post-fire tree regeneration processes.

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